F23PP469-VS
VARIABLE SPEED DRIVE

SYSTEM INTERCONNECTION DIAGRAM DRG. No. 2000-EL005-B02-10002

11,000V

RD WH BU

FCF 160A 160A 160A

QAC 400A

BCT1 50 / 1A 2.5VA 5P20 — 1S1 1S2

BCT2 50 / 1A 2.5VA 5P20 — 1S1 1S2

BCT3 50 / 1A 2.5VA 5P20 — 1S1 1S2

SWITCHBOARD EARTH BAR — NOTE 3

SUB FLOOR DROP BOX

BCN 50 / 1A 1.5VA 10P10 — S1 S2

BAD

PFV1 — RD RD RD

QCE

EARTH BAR

TRANSFORMER EARTH BAR

X4 1 X4 2 X4 3

X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13

2000-EL003-F16-10011 SHEET 14

L1 L2 L3 — LINE SIDE INCOMING SUPPLY SHEET 6

OUTGOING MOTOR SUPPLY SHEET 6

T1 T2 T3

CABLE SCREEN EARTHED

M 3~

F23-PP-469
FINAL TAILS PUMP 4
3,300V
800 kW

MOTOR SPACE HEATER CIRCUIT SHEET 6

X2 54 53

F23PP469HT-L — 1 (H1) — 0.3kW — ANTI-CONDENSATION HEATER

F23PP469HT-N — 3 (H2)

F23-PP-469
MOTOR HEATER TERMINAL BOX
2000-ME063B-F14-10013

A80 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6

X81

PT100 RTD INPUT CH1	+ 13	F23PP469A-RD	MOTOR WINDING RTD (RD PHASE) (TD1)
	COM 14	F23PP469A-WH	
	- 15	F23PP469A-BK	
PT100 RTD INPUT CH2	+ 16	F23PP469B-RD	MOTOR WINDING RTD (WH PHASE) (TD3)
	COM 17	F23PP469B-WH	
	- 18	F23PP469B-BK	
PT100 RTD INPUT CH3	+ 19	F23PP469C-RD	MOTOR WINDING RTD (BU PHASE) (TD5)
	COM 20	F23PP469C-WH	
	- 21	F23PP469C-BK	
PT100 RTD INPUT CH4	+ 22	F23PP469D-RD	MOTOR BEARING RTD (DRIVE END) (BD10)
	COM 23	F23PP469D-WH	
	- 24	F23PP469D-BK	
PT100 RTD INPUT CH5	+ 25	F23PP469E-RD	MOTOR BEARING RTD (NON DRIVE END) (BD20)
	COM 26	F23PP469E-WH	
	- 27	F23PP469E-BK	
PT100 RTD INPUT CH6	+ 28		
	COM 29		
	- 30		
PT100 RTD INPUT CH7	+ 31		
	COM 32		
	- 33		
PT100 RTD INPUT CH8	+ 34		
	COM 35		
	- 36		

RTD

F23-PP-469
MOTOR RTD TERMINAL BOX
2000-ME063B-F14-10013

FIELD CABLE TERMINALS

F23-SW-016 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10011 SHEET 7 3A & 3B

X3 1139 3 — X2 67

X3 3213 17 — X2 68 — K41 — K41 — MV CONTACTOR ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10011 SHEET 10 4A

X2 1179 4 — X2 63 — K41 — K41 — K41

X2 1199 25 — X2 64 — MV INPUT ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD TRIP 2000-EL003-F16-10011 SHEET 10 3A

X2 1219 3 — X2 65

X2 1239 24 — X2 66 — K41 — K41 — K41 — MV INPUT TRIP CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10011 SHEET 8 5A & 5D

X2 1259 56 — X2 70

X2 1279 44 — X2 69

A30 (PART OF) — X2 15 — TRIP — X2 16 — ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F23-DB-101
2000-F23-TLD-EL-10054

RD WH BU BK — R6 W6 B6 — 25A — 415VAC — EARTH BAR

F23DB101-R6 — X4 1
F23DB101-W6 — 2
F23DB101-B6 — 3
F23DB101-BK6 — N
EARTH BAR — PE1

VSD AUXILIARY POWER SUPPLY SHEET 6

F23-DB-201
2000-F23-TLD-EL-10101

RD BK — Q176 — 10A — 240VAC — EARTH BAR

F23DB201-Q176 (RD) — X4 5
F23DB201-Q176 (BK) — 6
EARTH BAR — PE2

VSD UPS POWER SUPPLY SHEET 6

TO FINAL TAILS PUMPS
415V SWITCHROOM
F23-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F23-ETH-001
2000-F23-TER-IC-10003

CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6

X47 — ETHERNET ADAPTOR RJ45 — SHEET 6

A2 (PART OF) — DI-2B 9

SAFE TORQUE OFF CIRCUIT SHEET 6

X86

F23PP469VS-1409 — X1 1 — LOS
F23PP469VS-1419 — 2
F23PP469VS-1429 — 3
F23PP469VS-1439 — 4

X2 13 — F23PP469VS-1459 — X1 5 — START
24VDC CONTROL SHEET 6 — 14 — F23PP469VS-1469 — 6
E

F23PP469-LCS
LOCAL CONTROL STATION

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F23PP469-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. IF THE CABLE SCREENS PASS THROUGH CORE BALANCE CT BEFORE BREAKING OUT THEN THE SCREEN EARTHS MUST BE BOUGHT BACK THROUGH THE CORE BALANCE CT AND TERMINATED TO EARTH.

LEGEND

- ⊗ 11kV SWITCHBOARD TERMINAL
- ⊗ VSD TERMINAL
- ◒ FIELD TERMINAL
- ◑ DCS OR I/O TERMINAL
- ◧ MARSHALLING TERMINAL
- (x) DEVICE TERMINAL

PANEL A06

2000-F23-SCM-EL-10111_Rev1

F23-SW-016 - F23-PP-469 final tails pump 4 VSD feeder - schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SCM-EL-10111.DWG 020CT25 9:03AM HOTSEAT04